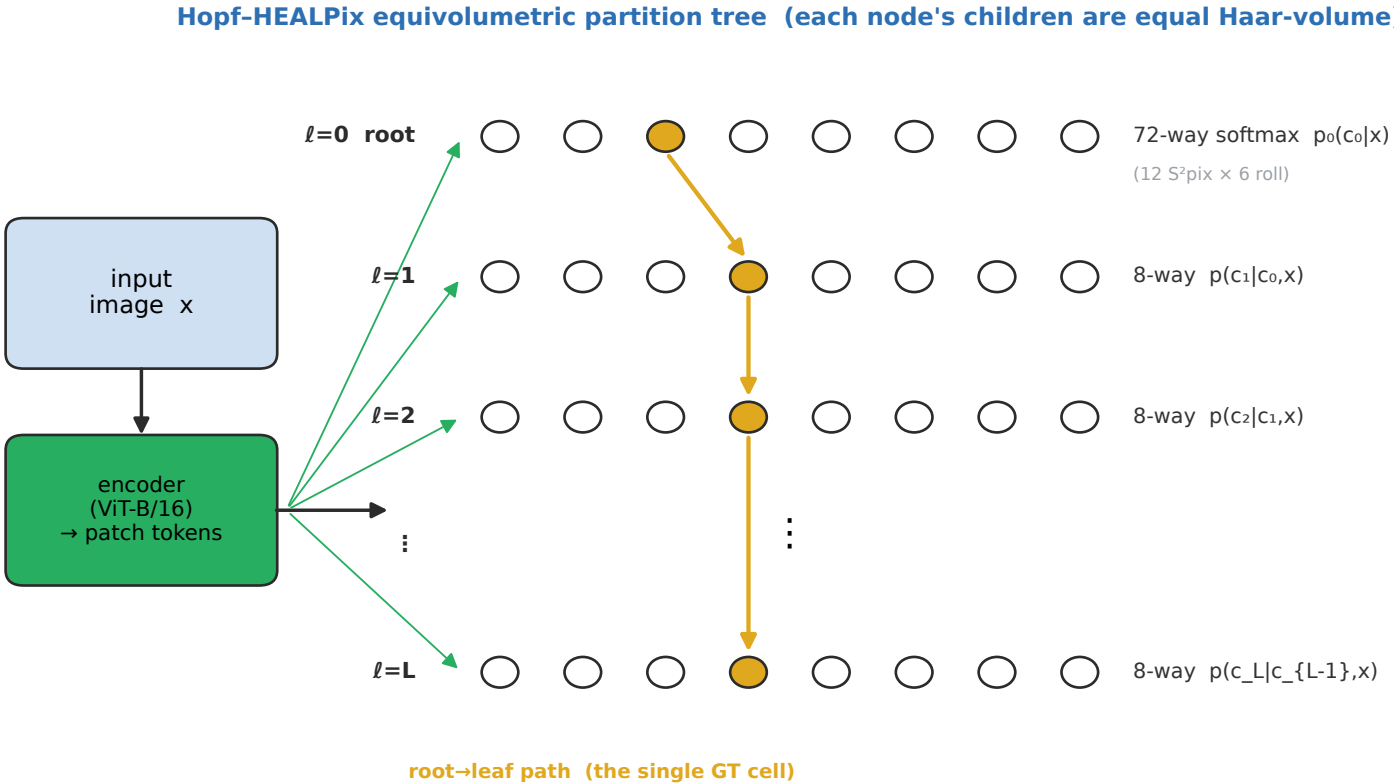
NeST: a tree-structured softmax over an equivolumetric partition is an exactly-normalized density



density = path-product × N/V

$$\log p(R|x) = \log p_0(c_0) + \Sigma_l \log p(c_l|\cdot) + \log(N / V)$$

- **Exact normalization**

  $\int p\, dR = 1$ by construction
  (each softmax sums to 1 over an equal-volume partition)

- **Exact uniform**

  uniform logits ⇒ $p \equiv 1/V$
  $= -\ln(\pi^2)$, no grid residual

- **O(L) evaluation**

  score only L+1 ≈ 7 nodes
  vs full-grid ~2.36M cells